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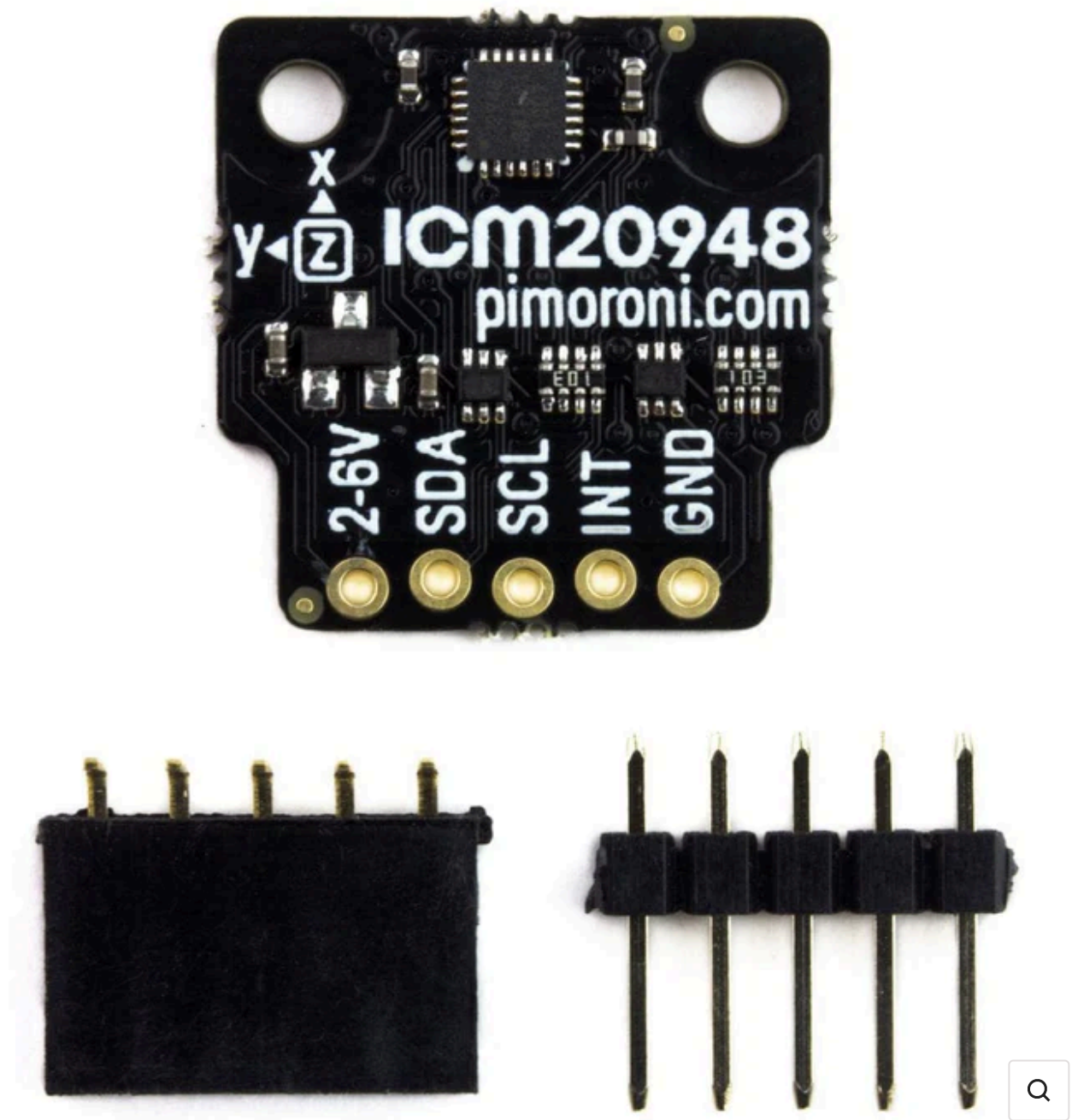


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ICM20948 9DoF Motion Sensor Breakout

PIMORONI SKU: PIM448

★★★★★ (1 Review)

£15 incl. VAT

Only 8 units left

- 1 +

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Compatible with:

RPi 3B

RPi 4B

RPi 5

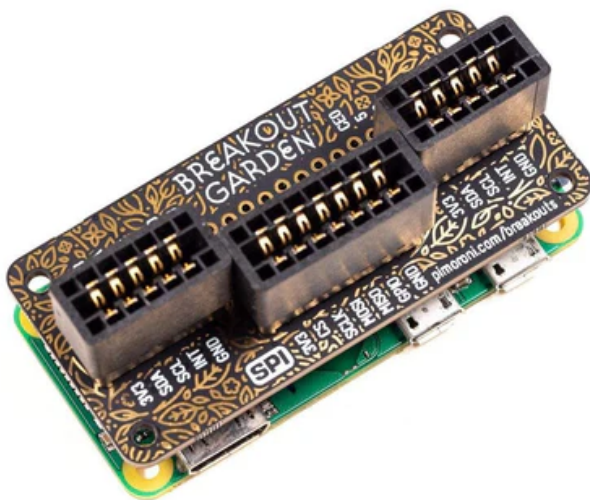
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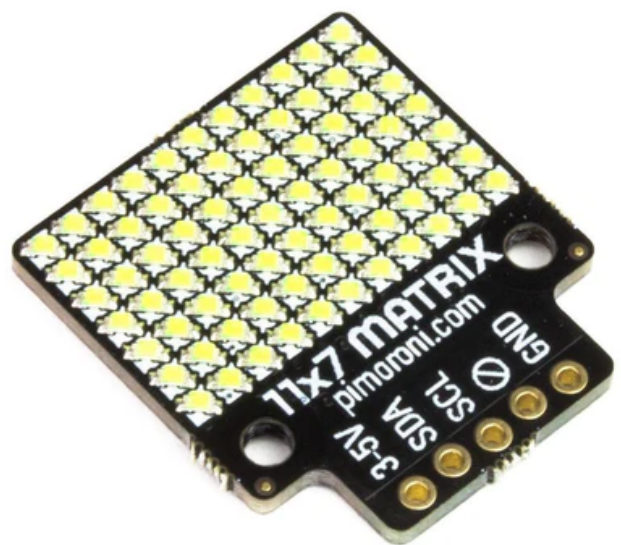
RPi Zero 2

Awesome Extras



★★★★★ (2 Reviews)

PIMORONI

Breakout Garden Mini (I2C + SPI)**£10.80** incl. VAT[NOTIFY ME](#)

PIMORONI

11x7 LED Matrix Breakout**£12.60** incl. VAT[NOTIFY ME](#)

This ICM20948 motion sensor is the bee's knees! Detect movement in nine different ranges of motion: 3-axis acceleration, 3-axis gyroscopic motion, and 3-axis compass heading.

[View our full Breakout Garden range here!](#)

Use this sensor in your DIY robot or drone builds to measure three different types of motion in three different axes accurately with 16-bit resolution. It's really low-power too, so it's ideal for embedding in battery-powered projects, and it's Raspberry Pi and Arduino-compatible.

It's compatible with our fancy [Breakout Garden HAT](#) (and [pHAT](#)), where using breakouts is as easy just popping it into one of the six slots and starting to grow your project, create, and code.

Features

- ICM20948 9DoF motion sensor ([datasheet](#))
- $\pm 2/\pm 4/\pm 8/\pm 16$ g 3-axis accelerometer
- $\pm 250/\pm 500/\pm 1000/\pm 2000$ DPS (degrees per second) 3-axis gyroscope
- 3-axis compass with wide range up to $\pm 4900 \mu\text{T}$
- 16-bit data output
- I2C interface (address 0x68/0x69 (cut trace))
- 3.3V or 5V compatible
- Reverse polarity protection
- Compatible with all models of Raspberry Pi, and Arduino
- [Python library](#)

Kit includes

- ICM20948 9DoF breakout
- 1x5 male header
- 1x5 female right-angle header

We've designed this breakout board so that you can solder on the piece of right-angle female header and pop it straight onto the bottom left 5 pins on your Raspberry Pi's GPIO header (pins 1, 3, 5, 7, 9)

Software

We've put together a [Python library](#) that you can use to read data from your ICM20948 9DoF Breakout, and an easy one-line installer to install everything.

Our software does not support Raspbian Wheezy.

Notes

Dimensions: 19 x 19 x 2.75mm

[View less](#)

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Payment methods



Your payment information is processed securely. We do not store credit card details nor have access to your credit card information.

Accreditations



5.00

Based on 1 review

[WRITE REVIEW](#)

Product Reviews

JH

 Verified Customer

John H

Saint Leonards-on-Sea, United Kingdom

ICM20948 9DoF Motion Sensor Breakout

Not as cheap as some units but seems to be stable and accurate. The magnetometer element of the 9 DoF is useful.

 2 people found this review helpful.

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1 year ago



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